

Safety Data Sheet

According to Notification of Thailand's Ministry of Industry B.E. 2555 (2012)



MACH 5 SM 20W-50

Version 0.0

Release Date: 2017/01/13

1. PRODUCT AND COMPANY IDENTIFICATION

Product name	: MACH 5 SM 20W-50
Other means of identification	: None
Recommended use	: Engine oil
Limitation of use	: This product should not be used for purposes other than those specified without the advice of an expert
Supplier	: PETRONAS International Marketing (Thailand) Co., Ltd
Registered corporate address	: 388 Exchange Tower, Unit Nos. 4101-4104 41 st floor, Sukhumvit Road, Klongtoey, Khet Klongtoey, Bangkok 10110
Emergency phone numbers	: +662-663 5881-5

2. HAZARDS IDENTIFICATION

GHS Classification

Skin corrosion/irritation	: Category 3
Serious eye damage/eye irritation	: Category 2A
Chronic aquatic toxicity	: Category 3

GHS Label element

Hazard pictograms



Signal word	: Warning
Hazard statements	: H316 Causes mild skin irritation. H319 Causes serious eye irritation. H412 Harmful to aquatic life with long lasting effects.
Precautionary statements	: Prevention: P264 Wash skin thoroughly after handling. P273 Avoid release to the environment. P280 Wear eye protection/ face protection. Response: P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P332 + P313 If skin irritation occurs: Get medical advice/ attention. P337 + P313 If eye irritation persists: Get medical advice/ attention. Disposal: P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

No information available.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Hazardous components

Chemical Name	CAS-No.	Concentration (%)
Paraffinic Distillate, Heavy, Hydrotreated	64742-54-7	>= 30 - < 50
Distillates (petroleum), solvent-dewaxed heavy paraffinic	64742-65-0	>= 20 - < 30
Distillates (petroleum), solvent-dewaxed heavy paraffinic	64742-70-7	>= 10 - < 20
Zinc alkyl dithiophosphate	113706-15-3	>= 1 - < 3.5
Long chain alkaryl amine	36878-20-3	>= 1 - < 3.5
Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased	68784-26-9	>= 0.1 - < 1
Tetrapropenylphenol	74499-35-7	>= 0.1 - < 1

4. FIRST AID MEASURES

If inhaled	: If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a medical doctor.
In case of skin contact	: If on skin, rinse well with water. If on clothes, remove clothes. If skin irritation persists, call a medical doctor.
In case of eye contact	: Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist. Immediately flush eye(s) with plenty of water.
If swallowed	: Keep respiratory tract clear. Do NOT give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. If symptoms persist, call a medical doctor.
General advice	: Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
Most important symptoms and effects, both acute and delayed	: No information available.

5. FIREFIGHTING MEASURES

Unsuitable media	extinguishing	: High volume water jet
Specific firefighting methods	hazards during extinguishing	: Do not allow run-off from fire fighting to enter drains or water courses. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed off in accordance with local regulations.
Special protective equipment for firefighters		: Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
- Environmental precautions : Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities.
Prevent further leakage or spillage if safe to do so.
- Methods and materials for containment and cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Normal measures for preventive fire protection.
- Advice on safe handling : Do not breathe vapours/dust.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations.
Avoid contact with skin and eyes.
- Conditions for safe storage : Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Keep container tightly closed in a dry and well-ventilated place. Electrical installations / working materials must comply with the technological safety standards.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Paraffinic Distillate, Heavy, Hydrotreated	64742-54-7	TWA (Inhalable fraction)	5 mg/m ³	ACGIH
	Further information: Upper Respiratory Tract irritation, Not classifiable as a human carcinogen			
Distillates (petroleum), solvent-dewaxed heavy paraffinic	64742-65-0	TWA (Inhalable fraction)	5 mg/m ³	ACGIH
	Further information: Upper Respiratory Tract irritation, Not classifiable as a human carcinogen			
	Further information: Upper Respiratory Tract irritation, Exposure by all routes should be carefully controlled to levels as low as possible., Suspected human carcinogen			
Paraffinic Distillate, Light, Hydrotreated	64742-55-8	TWA (Inhalable fraction)	5 mg/m ³	ACGIH

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	Further information: Upper Respiratory Tract irritation, Not classifiable as a human carcinogen
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Personal protective equipment

Hand protection

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water.
Tightly fitting safety goggles.
Wear face-shield and protective suit for abnormal processing problems.

Skin and body protection : Impervious clothing.
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Hygiene measures : When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Liquid
Colour	: No data available
Odour	: Not relevant
Odour Threshold	: Not relevant
pH	: Not applicable
Flash point	: 228 °C Method: Cleveland open cup
Evaporation rate	: Not applicable
Upper explosion limit	: Not applicable
Lower explosion limit	: Not applicable
Relative vapour density	: Not applicable
Relative density	: Not applicable
Density	: 0.893 g/cm ³ (15 °C)
Solubility(ies)	
Water solubility	: Not applicable
Solubility in other solvents	: Not applicable
Partition coefficient: n-octanol/water	: Not applicable
Decomposition temperature	: Not applicable
Viscosity, dynamic	: Not applicable
Viscosity, kinematic	: 166.7 mm ² /s (40 °C) 18.5 mm ² /s (100 °C)

10. STABILITY AND REACTIVITY

Reactivity	: Hazardous polymerisation does not occur.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, sparks, flame and build-up of static electricity.
Incompatible materials	: Strong oxidizing agents.
Hazardous decomposition products	: Fumes, smoke, carbon monoxide

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Product:

Acute dermal toxicity	: Acute toxicity estimate: > 5,000 mg/kg Method: Calculation method
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Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute inhalation toxicity	: LC50 (Rat): > 5.53 mg/l
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute inhalation toxicity	: LC50 (Rat): > 5.53 mg/l
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute inhalation toxicity	: LC50 (Rat): > 5.53 mg/l
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg

Zinc alkyl dithiophosphate:

Acute oral toxicity	: Remarks: No data available
Acute inhalation toxicity	: Remarks: No data available
Acute dermal toxicity	: Remarks: No data available

Long chain alkaryl amine:

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute inhalation toxicity	: Remarks: No data available
Acute dermal toxicity	: LD50 (Rat): > 2,000 mg/kg

Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute dermal toxicity	: LD50 (Rabbit): > 4,000 mg/kg

Tetrapropenylphenol:

Acute oral toxicity	: Remarks: No data available
Acute inhalation toxicity	: Remarks: No data available
Acute dermal toxicity	: Remarks: No data available

Skin corrosion/irritation

Product:

Remarks	: May cause skin irritation and/or dermatitis.
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Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Species	: Rabbit
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Result : No skin irritation
Distillates (petroleum), solvent-dewaxed heavy paraffinic:
 Result : No skin irritation
Distillates (petroleum), solvent-dewaxed heavy paraffinic:
 Species : Rabbit
 Exposure time : 4 h
 Result : No skin irritation
Zinc alkyl dithiophosphate:
 Result : Skin irritation
Long chain alkaryl amine:
 Species : Rabbit
 Result : No skin irritation
Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:
 Species : Rabbit
 Result : No skin irritation
Tetrapropenylphenol:
 Result : Skin irritation

Serious eye damage/eye irritation

Product:

Remarks : Vapours may cause irritation to the eyes, respiratory system and the skin.

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Species : Rabbit
 Result : No eye irritation

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Result : No eye irritation

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Species : Rabbit
 Result : No eye irritation

Zinc alkyl dithiophosphate:

Result : Irreversible effects on the eye

Long chain alkaryl amine:

Species : Rabbit
 Result : No eye irritation

Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:

Species : Rabbit
 Result : No eye irritation

Tetrapropenylphenol:

Result : Eye irritation

Respiratory or skin sensitisation

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Exposure routes : Skin contact
 Result : No evidence of skin sensitization.
 Exposure routes : Inhalation
 Remarks : No data available

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Remarks : No data available

Germ cell mutagenicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Germ cell mutagenicity - : No data available
Assessment

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Germ cell mutagenicity - : No data available
Assessment

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Germ cell mutagenicity - : No data available
Assessment

Carcinogenicity

Product:

Carcinogenicity - Assessment : Not classifiable as a human carcinogen.

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Carcinogenicity - Assessment : Contains less than 3 % DMSO extract as measured by IP 346
'Determination of polycyclic aromatics in unused lubricating
base oils and asphaltene free petroleum fractions — Dimethyl
sulphoxide extraction refractive index method'

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Carcinogenicity - Assessment : Contains less than 3 % DMSO extract as measured by IP 346
'Determination of polycyclic aromatics in unused lubricating
base oils and asphaltene free petroleum fractions — Dimethyl
sulphoxide extraction refractive index method'

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Carcinogenicity - Assessment : Contains less than 3 % DMSO extract as measured by IP 346
'Determination of polycyclic aromatics in unused lubricating
base oils and asphaltene free petroleum fractions — Dimethyl
sulphoxide extraction refractive index method'

Reproductive toxicity

Product:

Reproductive toxicity - : No toxicity to reproduction
Assessment

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Reproductive toxicity - : Not suspected to be a reproductive toxicant.
Assessment

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Reproductive toxicity - : Not suspected to be a reproductive toxicant.
Assessment

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Reproductive toxicity - : Not suspected to be a reproductive toxicant.
Assessment

Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:

Reproductive toxicity - : Some evidence of adverse effects on sexual function and
Assessment fertility, and/or on development, based on animal experiments.

Tetrapropenylphenol:

Reproductive toxicity - : Suspected human reproductive toxicant.
Assessment

STOT - single exposure

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Assessment : No specific organ toxicity following single exposure.

STOT - repeated exposure

Product:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Assessment : No specific organ toxicity following repeated exposure.

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Assessment : No specific organ toxicity following repeated exposure.

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Assessment : No specific organ toxicity following repeated exposure.

12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Toxicity to fish : LL50 (Pimephales promelas (fathead minnow)): > 100 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available

Toxicity to algae : NOEL (Pseudokirchneriella subcapitata (green algae)): >= 100 mg/l
Exposure time: 72 h

Toxicity to fish (Chronic toxicity) : NOELR (Oncorhynchus mykiss (rainbow trout)): >= 1,000 mg/l
Exposure time: 14 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEL (Daphnia magna (Water flea)): 10 mg/l
Exposure time: 21 d

Toxicity to microorganisms : NOEL: > 1.93 mg/l
Exposure time: 4 d

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Toxicity to fish : (Fish): > 100 mg/l

Toxicity to daphnia and other aquatic invertebrates : (Crustaceans): > 100 mg/l

Toxicity to algae : (Algae): > 100 mg/l

Toxicity to fish (Chronic toxicity) : Remarks: No data available

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : > 1 mg/l

Toxicity to microorganisms : Remarks: No data available

Toxicity to fish : (Fish): > 100 mg/l

Toxicity to daphnia and other aquatic invertebrates : (Crustaceans): > 100 mg/l

Toxicity to algae : (Algae): > 100 mg/l

Toxicity to fish (Chronic toxicity) : Remarks: No data available



Toxicity to daphnia and other aquatic invertebrates : > 1 mg/l

(Chronic toxicity)

Toxicity to microorganisms : Remarks: No data available

Zinc alkyl dithiophosphate:

Ecotoxicology Assessment

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

Long chain alkaryl amine:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): > 100 mg/l

Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l

Exposure time: 48 h

Toxicity to algae : EC50 (Desmodesmus subspicatus (green algae)): > 100 mg/l

Exposure time: 72 h

Toxicity to fish : Remarks: No data available

(Chronic toxicity)

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available

(Chronic toxicity)

Toxicity to microorganisms : IC50 (Activated sludge): > 100 mg/l

Exposure time: 3 h

Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:

Toxicity to fish : LL50 (Pimephales promelas (fathead minnow)): > 1,000 mg/l

Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): > 1,000 mg/l

Exposure time: 48 h

Toxicity to algae : EL50 (Pseudokirchneriella subcapitata (green algae)): > 500 mg/l

Exposure time: 96 h

Toxicity to fish : Remarks: No data available

(Chronic toxicity)

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available

(Chronic toxicity)

Toxicity to microorganisms : EC50 (Activated sludge): > 10,000 mg/l

Exposure time: 3 h

Ecotoxicology Assessment

Chronic aquatic toxicity : May cause long lasting harmful effects to aquatic life.

Tetrapropenylphenol:

Toxicity to fish : Remarks: No data available

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available

Toxicity to algae : Remarks: No data available

Toxicity to fish : Remarks: No data available

(Chronic toxicity) : Remarks: No data available

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available

(Chronic toxicity)

Toxicity to microorganisms : Remarks: No data available

Ecotoxicology Assessment

Acute aquatic toxicity : Very toxic to aquatic life.

Chronic aquatic toxicity : Very toxic to aquatic life with long lasting effects.

Persistence and degradability

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Biodegradability : Exposure time: 28 d
Result: Not readily biodegradable.Method: OECD Test
Guideline 301

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Biodegradability : Result: Inherently biodegradable.

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Biodegradability : Result: Inherently biodegradable.

Long chain alkaryl amine:

Biodegradability : Result: Not biodegradable

Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:

Biodegradability : Result: Not readily biodegradable.

Tetrapropenylphenol:

Biodegradability : Remarks: No data available

Bioaccumulative potential

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Bioaccumulation : Remarks: No data available

Partition coefficient: : Remarks: No data available
n-octanol/water

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Bioaccumulation : Remarks: Potentially bio-accumulative.

Partition coefficient: : log Pow: 2 - 6

n-octanol/water

Distillates (petroleum), solvent-dewaxed heavy paraffinic:

Bioaccumulation : Remarks: Potentially bio-accumulative.

Partition coefficient: : log Pow: >= 4

n-octanol/water

Long chain alkaryl amine:

Bioaccumulation : Species: Cyprinus carpio (Carp)
Bioconcentration factor (BCF): 1,730
Remarks: see user defined free text

Partition coefficient: : log Pow: 7.5

n-octanol/water

Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:

Bioaccumulation : Bioconcentration factor (BCF): 2.2

Partition coefficient: : log Pow: 9.5

n-octanol/water

Tetrapropenylphenol:

Bioaccumulation : Remarks: No data available

Partition coefficient: : Remarks: No data available

n-octanol/water

Mobility in soil

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Mobility : Medium: Soil
Remarks: No data available

Other adverse effects

Product:

Additional ecological : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
information Harmful to aquatic life with long lasting effects.

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Additional ecological : No data available
information

Distillates (petroleum), solvent-dewaxed heavy paraffinic:No data available
Distillates (petroleum), solvent-dewaxed heavy paraffinic:No data available
Zinc alkyl dithiophosphate:No data available
Long chain alkaryl amine:No data available
Phenol, dodecyl-, sulfurized, carbonates, calcium salts, overbased:No data available
Tetrapropenylphenol:No data available

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not dispose of waste into sewer.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.

14. TRANSPORT INFORMATION

International Regulation

UNRTDG

Not regulated as a dangerous good

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC

Code Not applicable for product as supplied.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

CH INV : Not in compliance with the inventory
TSCA : Not On TSCA Inventory
DSL : This product contains the following components that are not on the Canadian DSL nor NDSL.



	Acrylic copolymer	
	Interchangeable neutral oils	
	Tetrapropenylphenol	74499-35-7
AICS	: Not in compliance with the inventory	
NZIoC	: Not in compliance with the inventory	
ENCS	: Not in compliance with the inventory	
ISHL	: Not in compliance with the inventory	
KECI	: Not in compliance with the inventory	
PICCS	: Not in compliance with the inventory	
IECSC	: Not in compliance with the inventory	

16. OTHER INFORMATION

Revision Date : 2017/01/13

Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists
AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
CCHC	-	Chemicals Classification and Hazard Communication
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
ERG	-	Emergency Response Guide
GHS	-	Globally Harmonized System
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication
IECSC	-	Inventory of Existing Chemical Substances in China
IMDG	-	International Maritime Dangerous Goods
IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)
ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships
MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration



NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSPP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature
SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System

Disclaimer

To the best of PETRONAS knowledge, the information contained herein is accurate and reliable as of the date of publication. Any use of information provided should be undertaken in consultation with a professional with the appropriate expertise, as necessary in the circumstances. No warranty, express or implied, is given as to the quality, accuracy, reliability, applicability or completeness of the contents of this SDS. The information presented here pertains only to the product as shipped. It is the responsibility of the customer to ensure that any activities relating to the product comply with all federal, state or local laws. Any hazards associated with any product regulatory requirements are subject to change and may differ between various locations. Except to the extent required by law, re-publication or retransmission of this SDS, in whole or in part, is strictly prohibited. PETRONAS does not take responsibility for use, transportation, storage, handling, packaging or disposal of the product that is beyond our knowledge and recommendation by this SDS.

Product Stewardship Advisory:

PETRONAS aims to increase awareness of all the hazards associated with the storage, handling and use of its products. Thoroughly reviewing the accompanying Safety Data Sheets and disseminating the information to all dependent and interested parties is an essential part of any 'Responsible Care' programme.

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